EgoQuant

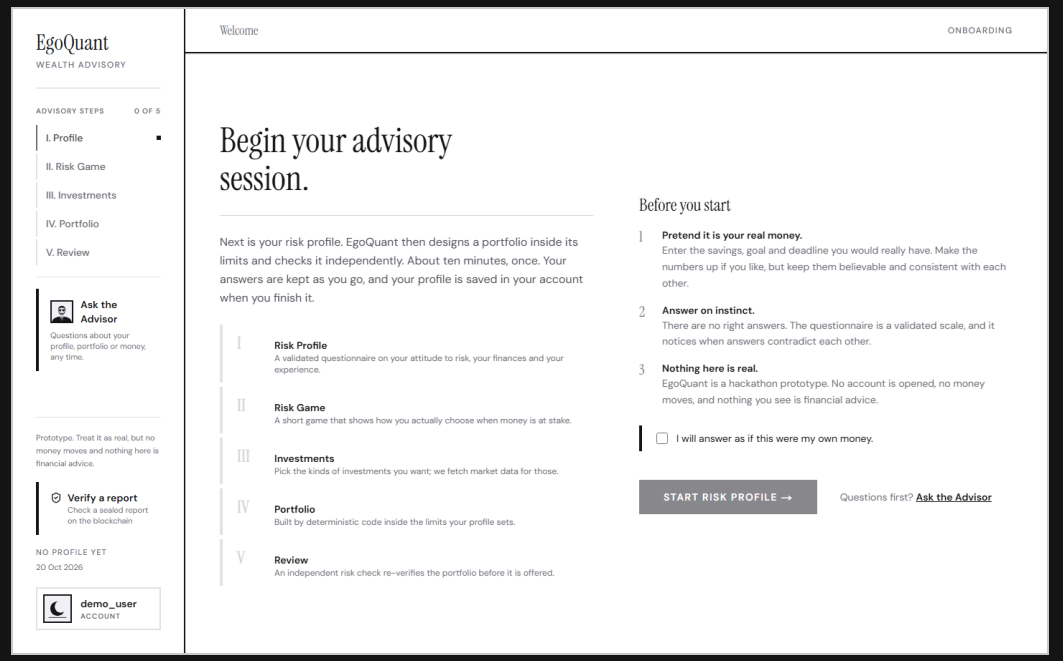
Your portfolio should understand you

A behavioural approach to wealth advisory: a wealth-advisory agent you can check

2026 Shenzhen FinTechathon · WeBank International Track · Topic B: Wealth Advisory Agent

Team EgoQuant · NYU Abu Dhabi · 20 October 2026

Live: fintechaton-frontend.vercel.app



The problem a retail investor has

Most apps show a portfolio and ask for trust. Three things the investor cannot do today:

1

Judge fit

Is this portfolio suitable for my finances and for how I really behave under loss?

2

See why

Which rule set this limit, where does this number come from, and who checked it?

3

Prove later

Can I show afterwards, unchanged, what I was told?

EgoQuant makes three promises in return: Fit, Proof, Trace.

The one rule that organises the system

Numbers that matter come from deterministic, tested code. AI only explains. An independent Risk Engine has the final word.

1

Code decides

Scores, the risk category, every limit, the portfolio weights, approvals and trade verdicts. No language model can write to any of them.

2

AI explains

The advisor explains the engine's facts, can run two whitelisted tools (what-if, check one trade), and every number in its answer is checked against the engine.

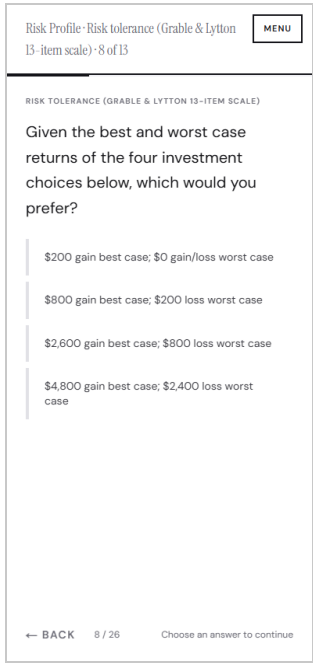
3

One labelled exception

The news briefing's tone label for a headline and a fund comes from a model. It never sets a number the person acts on.

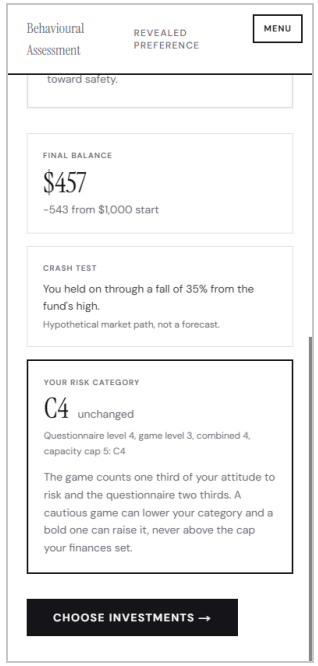
The Risk Engine re-computes every limit from the weights alone and fails closed on anything unknown.

The journey: about ten minutes, five steps



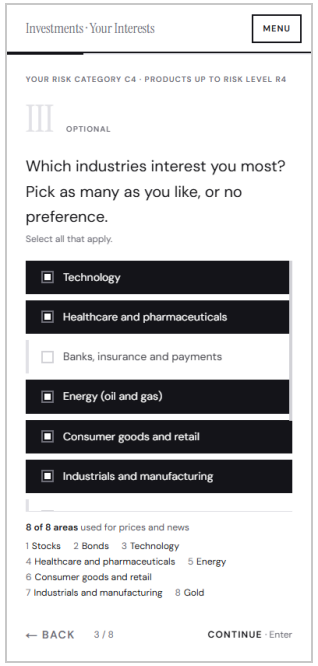
I. Profile

25 questions; two published scales quoted verbatim; a fixed program scores them



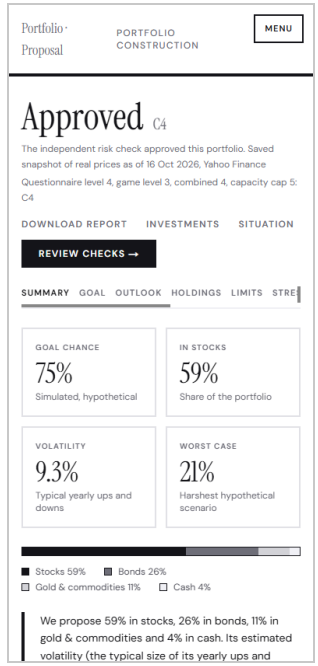
II. Risk Game

Seven rounds and a crash test: how the person behaves, not only what they say



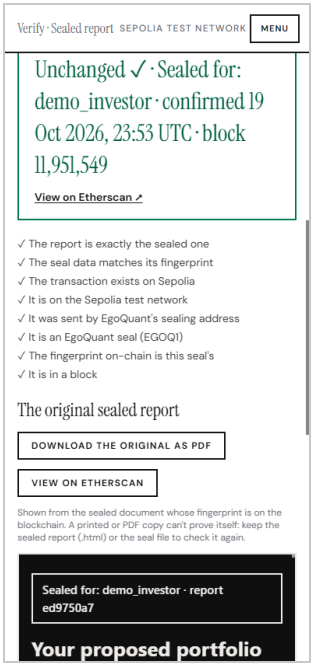
III. Investments

Only what the category allows can be ticked; greyed-out options say why



IV. Portfolio

Approved by the Risk Engine; the goal chance as N in 100 simulated futures



V. Review and seal

The independent checks, then Confirm and seal, then Verify a report

Suitability by design: the lower of willing and able

- Category C1 (most careful) to C5 = the LOWER of capacity (what the finances allow) and attitude to risk
- Attitude = two thirds questionnaire + one third Risk Game; a bold game can never pass the capacity cap
- "I could not accept any loss" always gives C1
- Attitude: Grable and Lytton's 13-item scale, verbatim. Knowledge: Lusardi and Mitchell's Big Three; it restricts complex products, never the category
- Capacity averages horizon, savings, share of wealth and income; money that must come back and a short horizon are hard caps
- Product risk levels R1 to R5: a Cn client may hold up to Rn

Category	Product level	Stocks	Crypto	Loss limit in a severe fall
C1	R1	0%	0%	5%
C2	R2	0%	0%	10%
C3	R3	40%	0%	20%
C4	R4	70%	0%	35%
C5	R5	90%	5%	45%

Provisional placeholders in one configuration file; a bank replaces them with its approved table. Every report carries a DRAFT LIMITS banner.

Six specialised parts, one independent check

Questionnaire + Risk Game

category C1 to C5, numeric limits (deterministic code)

↓

Agent C · market intelligence

allowed universe per category and platform; real prices (committed snapshot of 292 tickers, about 19 years); data provenance

→

Agent A · quant

about 100 candidate portfolios (20 risk points x 5 styles) inside every hard limit; goal simulation on 10,000 futures

←

Agent B · behavioural twin

IBKR trade history, crash test; returns limits that can ONLY tighten

↓

Agent D · news briefing

headlines per holding, read-only; the one labelled AI tone label

Risk Engine · final word

re-computes every limit from the weights alone; unknown product = fail closed; only an approved portfolio is shown

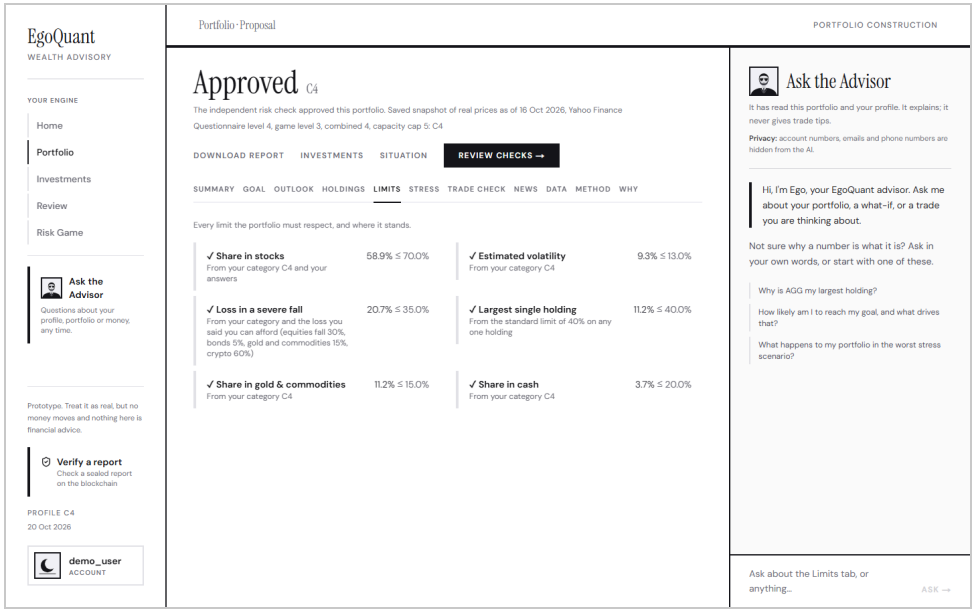
Advisor (OpenAI)

explains sealed facts; tools: what-if, check one trade; number grounding, redaction, compliance guard

Next.js on Vercel → FastAPI on Railway → Supabase (Auth + Postgres with Row-Level Security) · OpenAI (explanations only) · Sepolia test chain (fingerprints only)

Behaviour counts, and can only make it safer

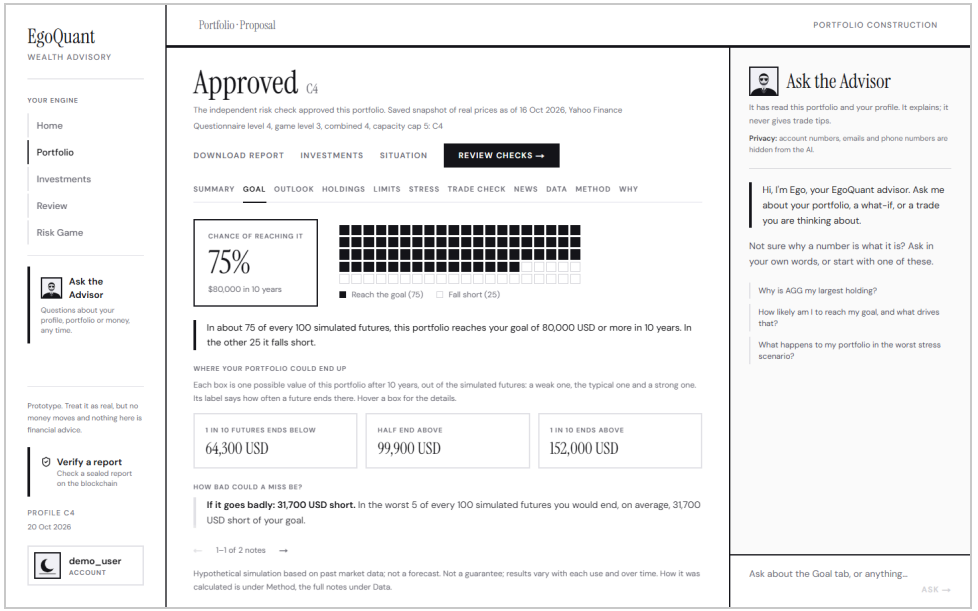
- The Risk Game gives every person a behavioural signal; it counts one third of the attitude to risk and never pushes past the capacity cap
- A real Interactive Brokers history (read-only upload or one-off Flex token; nothing stored) is stronger evidence and tightens only
- Demo: the balanced investor, category C3. Without a history: 23% stocks, 87 of 100 simulated futures reach the goal
- With an invented history that sold half its stocks in two market falls: the loss limit moves from 20% to 8.8%, stocks from 23% to 8%, the goal chance to 86 of 100; the Limits tab shows the line and the report says why
- The eager beginner: C5 by the questionnaire, C4 after a cautious game; crypto stays closed because the knowledge questions were not answered correctly



The Limits tab (screenshot: a different example, category C4): every limit with its source, including the behavioural limit from the person's own trades

A goal simulation that is honest about history

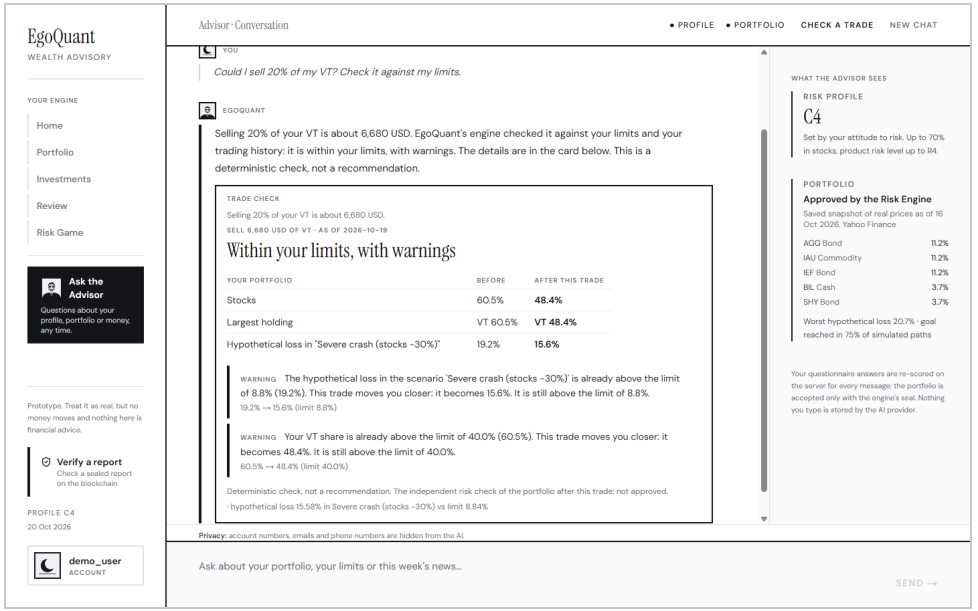
- The chance of reaching the goal is shown as "N in 100 simulated futures", never as a forecast
- 10,000 futures, the same for every candidate; month-long blocks of real daily returns, so crashes and rebounds stay clustered
- Regime-aware sampler: a four-state Hidden Markov model fitted offline, loaded as a hash-checked file, with a fall-back and a kill switch
- Two bias corrections: bonds and cash get today's rate, not their past average; stocks keep every real move but get the average their market sensitivity earned
- Our own calibration backtest (rolling start dates, refitted on earlier data only): Brier score better than the plain bootstrap by 0.007 at five years and 0.011 at ten; on the later dates the gain is within the noise, and realised outcomes can differ from the simulated chance by tens of points. Hence "simulated", never "probable"



The Goal tab: the chance as 100 boxes, where the portfolio could end up, how bad a miss could be

The advisor can run the engine but cannot decide

- Two whitelisted tools: run a what-if (the real engine on a validated scenario) and check one trade (the deterministic trade check on the saved history); no field for a limit, a category or an identity; at most two calls per message
- Number grounding: every percentage, "N in 100", amount and multiplier must come from the engine's facts or the user's own words, otherwise the reply is regenerated once, then replaced by a safe fallback
- Redaction gateway: names, e-mails, phone numbers and account ids become placeholders before any model call; requests use store=false
- Compliance guard: 36 patterns (English and Chinese) withhold anything that reads like a trade instruction, a price target, a guarantee or a forecast
- Live red-teams against the real model (raw results in eval/results): earlier prompts 19/19, 19/19, 21/21, 24/24. On 20 Oct the red-team of the final advisor found a real defect in the trade tool (the model filled two size fields, so no trade check completed: a safe failure), fixed within the hour as advisor-v11; the re-run passed 41 of 42 cases (the one flag is a false alarm of our own check) and the trade explanation 51 of 51



A trade question in the chat: the verdict card under the reply is the real engine's check

Security and compliance: what we built and what we measured

3,554

automatic tests, offline; last complete run 3,538 of 3,540 passed, 1 failed (a timing test of the news page), 1 skipped; 14 added since ran green

71 of 71

read-only deploy checks passed on the live service (20 Oct)

0

known vulnerabilities in the 56 pinned packages (pip-audit, 20 Oct)

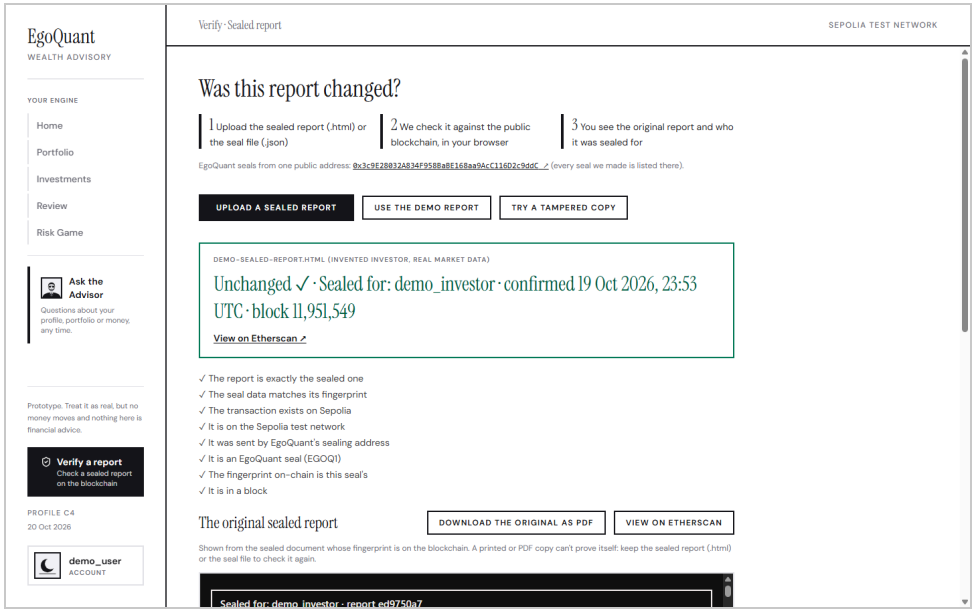
28

known gaps listed openly with owners, next to a threat table T1 to T19

- One person can never read another's data: every database call runs as the signed-in user under Row-Level Security on all 12 tables, the backend filters again, and there is no all-powerful key (the account store refuses to start with one); the RLS proof passed on the real project
- Less data, kept shorter: broker credentials never asked for, tokens used once, uploaded files parsed in memory, every row expires after 180 days, "delete my data" erases every table (a test fails if one is forgotten)
- Hostile input: strict schemas, no echo of what was sent, 80 + 33 + 129 hostile and extreme cases with no server error; size and rate limits; security headers and a content-security policy; secrets scanned on every push
- Fails safely: a slow price feed switches the whole build to a labelled snapshot; the engine works without the AI; kill switches for every optional part

Tamper-evident advice: the on-chain seal

- When a signed-in investor confirms an approved plan, the exact report is rendered once, marked "Sealed for: <username>", and a record of the advice is built
- Only a fingerprint goes on chain: EGOQ1 + SHA-256(salt ‖ record), 37 bytes in a 0-ETH transaction on Ethereum Sepolia (test network); nothing personal leaves the database
- Verify a report (no account needed) recomputes the fingerprint in the browser and reads the chain: "Unchanged ✓" or "Does not match ✗"; one changed character fails
- Real transactions: the Verify page's demo report (block 11,951,549); a live signed-in seal on 20 Oct (block 11,954,258); sealing address 0x3c9E…9ddC; the evidence list names every seal with its Etherscan link
- What it proves: this exact report existed, unchanged, no later than that block, for that username. What it does not: that the advice was right. Older seals still verify; "delete my data" removes the record and its salt. Roadmap: the same format on FISCO BCOS



Verify a report: a real sealed report checked against the live chain, every check passed

Honest limits, and what we would do next

What it is not, and where it is weak

- A licensed or regulated service: the category limits are placeholders, nothing has been reviewed by a regulator
- The Risk Game has no psychometric validation; its effect is bounded and it has kill switches
- The simulation replays about 19 years of history; it is not a probability
- Prices: a Yahoo Finance snapshot accepted for the competition only; the live site runs on saved real prices
- English only; Interactive Brokers only; the blockchain is a test network
- No third-party penetration test; the newest advisor prompt not yet live red-teamed

Roadmap, in order

- Institutional calibration of the category table; periodic suitability review
- Validate the Risk Game on a real sample; make the result of record non-replayable
- Licensed market data, FX and tax awareness
- The same salted fingerprint on FISCO BCOS, with a countersigning authority
- Chinese-language advice with its own reviewed guard; more brokers
- Third-party penetration test, monitoring, a per-user AI spending cap

Try it yourself

Live site: fintechaton-frontend.vercel.app · API and interactive docs: web-production-ee5cc.up.railway.app/docs

Two minutes, no account: Verify a report → Use the demo report → Try a tampered copy

Ten minutes with the demo account: profile, Risk Game, investments, portfolio, trade check, confirm and seal, verify

Documents 01 to 09 (summary, security and compliance, methods, testing and evidence, user guide, limitations, judges' questions, API reference, on-chain seal); code at the tag submission-20oct in both repositories

Every limit is a proposal, every simulation is hypothetical, and every claim names a file, a test or a measured number. Thank you.